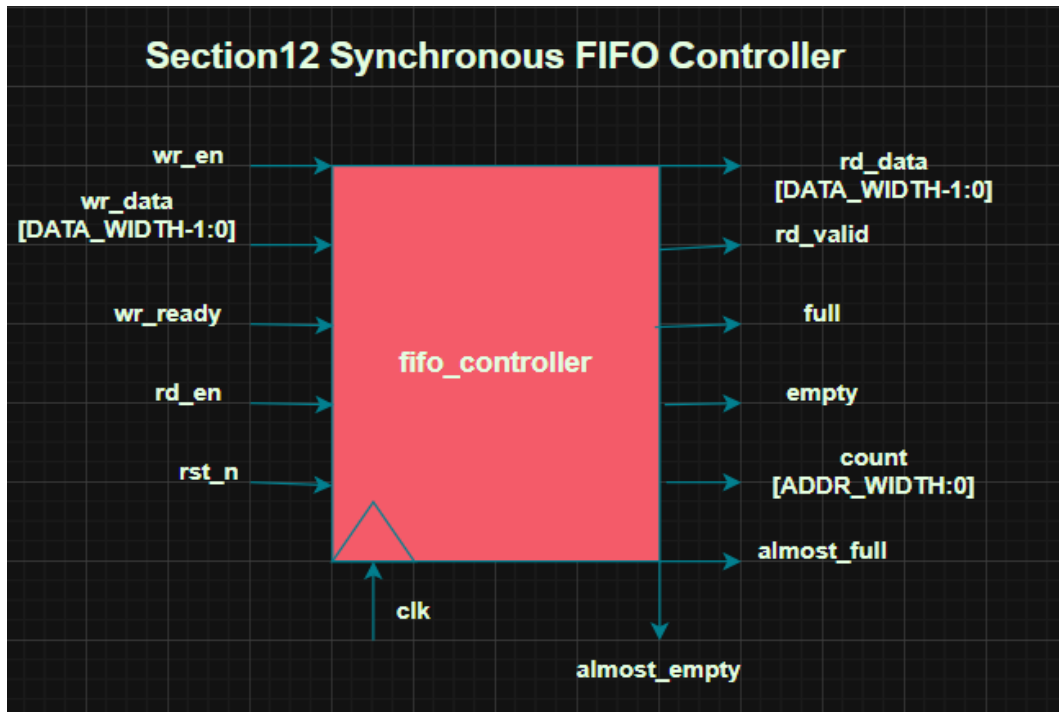


NAME: MUHAMMAD USMAN

REG. NO# 2024-EE-171

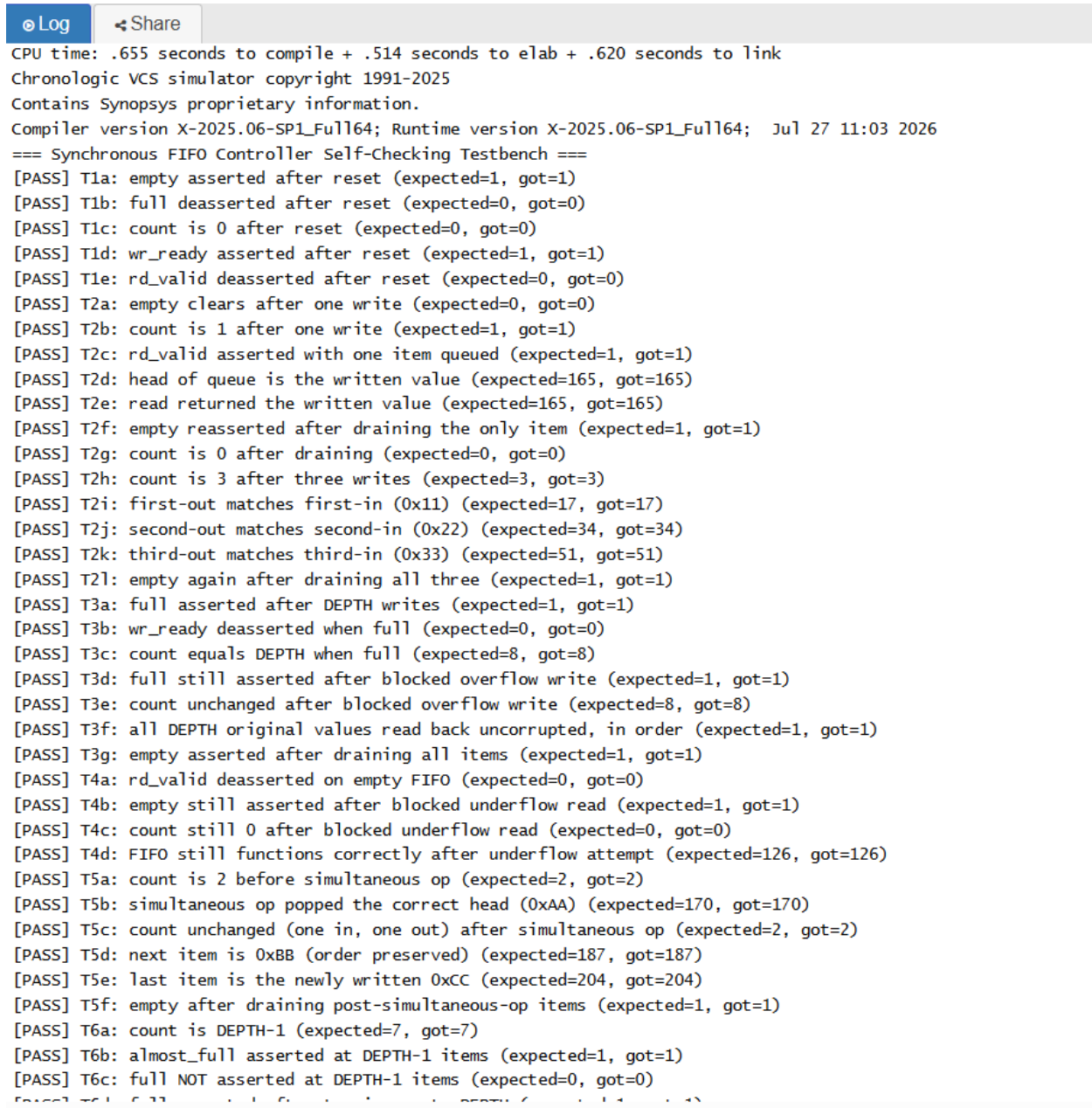
1-EDA Playground link: s12: <https://edaplayground.com/x/WS2r>

2-Schematic Diagram:



Section12_schematic_diagram

3-Output Screenshot:



```
Log Share
CPU time: .655 seconds to compile + .514 seconds to elab + .620 seconds to link
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164; Jul 27 11:03 2026
=== Synchronous FIFO Controller Self-Checking Testbench ===
[PASS] T1a: empty asserted after reset (expected=1, got=1)
[PASS] T1b: full deasserted after reset (expected=0, got=0)
[PASS] T1c: count is 0 after reset (expected=0, got=0)
[PASS] T1d: wr_ready asserted after reset (expected=1, got=1)
[PASS] T1e: rd_valid deasserted after reset (expected=0, got=0)
[PASS] T2a: empty clears after one write (expected=0, got=0)
[PASS] T2b: count is 1 after one write (expected=1, got=1)
[PASS] T2c: rd_valid asserted with one item queued (expected=1, got=1)
[PASS] T2d: head of queue is the written value (expected=165, got=165)
[PASS] T2e: read returned the written value (expected=165, got=165)
[PASS] T2f: empty reasserted after draining the only item (expected=1, got=1)
[PASS] T2g: count is 0 after draining (expected=0, got=0)
[PASS] T2h: count is 3 after three writes (expected=3, got=3)
[PASS] T2i: first-out matches first-in (0x11) (expected=17, got=17)
[PASS] T2j: second-out matches second-in (0x22) (expected=34, got=34)
[PASS] T2k: third-out matches third-in (0x33) (expected=51, got=51)
[PASS] T2l: empty again after draining all three (expected=1, got=1)
[PASS] T3a: full asserted after DEPTH writes (expected=1, got=1)
[PASS] T3b: wr_ready deasserted when full (expected=0, got=0)
[PASS] T3c: count equals DEPTH when full (expected=8, got=8)
[PASS] T3d: full still asserted after blocked overflow write (expected=1, got=1)
[PASS] T3e: count unchanged after blocked overflow write (expected=8, got=8)
[PASS] T3f: all DEPTH original values read back uncorrupted, in order (expected=1, got=1)
[PASS] T3g: empty asserted after draining all items (expected=1, got=1)
[PASS] T4a: rd_valid deasserted on empty FIFO (expected=0, got=0)
[PASS] T4b: empty still asserted after blocked underflow read (expected=1, got=1)
[PASS] T4c: count still 0 after blocked underflow read (expected=0, got=0)
[PASS] T4d: FIFO still functions correctly after underflow attempt (expected=126, got=126)
[PASS] T5a: count is 2 before simultaneous op (expected=2, got=2)
[PASS] T5b: simultaneous op popped the correct head (0xAA) (expected=170, got=170)
[PASS] T5c: count unchanged (one in, one out) after simultaneous op (expected=2, got=2)
[PASS] T5d: next item is 0xBB (order preserved) (expected=187, got=187)
[PASS] T5e: last item is the newly written 0xCC (expected=204, got=204)
[PASS] T5f: empty after draining post-simultaneous-op items (expected=1, got=1)
[PASS] T6a: count is DEPTH-1 (expected=7, got=7)
[PASS] T6b: almost_full asserted at DEPTH-1 items (expected=1, got=1)
[PASS] T6c: full NOT asserted at DEPTH-1 items (expected=0, got=0)
```

Task12_log_result_1.png

Log

Share

```
[PASS] T2j: second-out matches second-in (0x22) (expected=34, got=34)
[PASS] T2k: third-out matches third-in (0x33) (expected=51, got=51)
[PASS] T2l: empty again after draining all three (expected=1, got=1)
[PASS] T3a: full asserted after DEPTH writes (expected=1, got=1)
[PASS] T3b: wr_ready deasserted when full (expected=0, got=0)
[PASS] T3c: count equals DEPTH when full (expected=8, got=8)
[PASS] T3d: full still asserted after blocked overflow write (expected=1, got=1)
[PASS] T3e: count unchanged after blocked overflow write (expected=8, got=8)
[PASS] T3f: all DEPTH original values read back uncorrupted, in order (expected=1, got=1)
[PASS] T3g: empty asserted after draining all items (expected=1, got=1)
[PASS] T4a: rd_valid deasserted on empty FIFO (expected=0, got=0)
[PASS] T4b: empty still asserted after blocked underflow read (expected=1, got=1)
[PASS] T4c: count still 0 after blocked underflow read (expected=0, got=0)
[PASS] T4d: FIFO still functions correctly after underflow attempt (expected=126, got=126)
[PASS] T5a: count is 2 before simultaneous op (expected=2, got=2)
[PASS] T5b: simultaneous op popped the correct head (0xAA) (expected=170, got=170)
[PASS] T5c: count unchanged (one in, one out) after simultaneous op (expected=2, got=2)
[PASS] T5d: next item is 0xBB (order preserved) (expected=187, got=187)
[PASS] T5e: last item is the newly written 0xCC (expected=204, got=204)
[PASS] T5f: empty after draining post-simultaneous-op items (expected=1, got=1)
[PASS] T6a: count is DEPTH-1 (expected=7, got=7)
[PASS] T6b: almost_full asserted at DEPTH-1 items (expected=1, got=1)
[PASS] T6c: full NOT asserted at DEPTH-1 items (expected=0, got=0)
[PASS] T6d: full asserted after topping up to DEPTH (expected=1, got=1)
[PASS] T6e: almost_full deasserted once truly full (expected=0, got=0)
[PASS] T6f: count is 1 (expected=1, got=1)
[PASS] T6g: almost_empty asserted at 1 item (expected=1, got=1)
[PASS] T6h: empty NOT asserted at 1 item (expected=0, got=0)
```

=====

TOTAL: 42 passed, 0 failed

RESULT: ALL TESTS PASSED

=====

\$finish called from file "testbench.sv", line 227.

\$finish at simulation time 510000

V C S S i m u l a t i o n R e p o r t

Time: 510000 ps

CPU Time: 0.550 seconds; Data structure size: 0.0Mb

Mon Jul 27 11:03:52 2026

Finding VCD file...

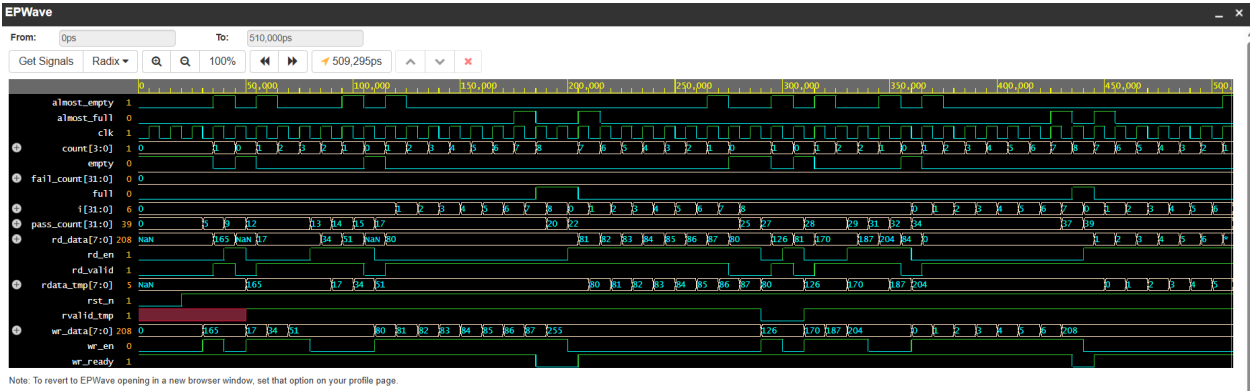
./dump.vcd

[2026-07-27 15:03:52 UTC] Opening EPWave...

Done

Task12_log_result.png

4-Waveform Screenshot:



Section12_waveform.png